

Study Report: IT Support / Helpdesk - Career Path & Core Skills

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Family:	IT Support / Helpdesk
Level:	Beginner to Advanced
Tracks:	Beginner, Intermediate, Advanced
Chapters:	18
Source:	Official docs, free libraries, and industry-standard references (see sources and books)

Official sources and free libraries

- <https://learn.microsoft.com/windows/>
- <https://learn.microsoft.com/microsoft-365/>
- <https://learn.microsoft.com/training/paths/az-900-describe-cloud-concepts/>
- <https://www.comptia.org/certifications/a>
- <https://ubuntu.com/tutorials/command-line-for-beginners>
- <https://www.axelos.com/certifications/itil-service-management/itil-4-foundation>

Summary

Enterprise IT support career path covering CompTIA A+ hardware and software concepts, structured troubleshooting, Windows and Linux administration, TCP/IP networking, ticketing systems, remote support, Active Directory basics, and ITIL service management. Built from Microsoft Learn, CompTIA objectives, and Ubuntu/Linux documentation for hands-on helpdesk work.

Skill stack by level

Recommended books

1. CompTIA A+ Complete Study Guide - Quentin Docter et al.

Level: Beginner | Hardware, OS, networking, and troubleshooting for helpdesk roles.

2. The Practice of System and Network Administration - Thomas Limoncelli et al.

Level: Intermediate | Enterprise IT operations, monitoring, and incident handling.

3. Windows Client Documentation - Microsoft Learn

Level: Beginner | Windows 11 administration, troubleshooting, and Intune basics.

4. ITIL 4 Foundation - AXELOS

Level: Beginner | IT service management framework used by global helpdesks.

5. The Linux Command Line - William Shotts

Level: Beginner | Free book for Linux shell, files, permissions, and scripting.

End-to-end projects

Project 1: Helpdesk Ticket Simulation Lab

Beginner | 1-2 weeks - Process 20 simulated tickets: triage, troubleshoot, document resolution, calculate SLA metrics.

- Triage: Triage spreadsheet
- Resolve: Resolution log
- KB: KB articles PDF
- Metrics: SLA dashboard

Project 2: Small Office IT Setup

Intermediate | 2-3 weeks - Design and document network for 15 users: IP plan, Active Directory, backup, security baseline.

- Network: Network diagram Visio/draw.io
- AD: AD structure doc
- Backup: Backup policy PDF
- Security: Security checklist signed off

Project 3: Incident Response Tabletop Exercise

Advanced | 3-4 weeks - Simulate ransomware incident: detect, contain, eradicate, recover, and write post-incident report.

- Plan: IR plan PDF
- Simulate: Exercise timeline
- Contain: Containment log
- Report: PIR document

Chapters

Track: Beginner

Chapter 1: IT Support Role and Service Desk Tiers [Beginner]

Chapter focus: IT Support Role and Service Desk Tiers** *(Level: Beginner)

IT support keeps employees productive by resolving hardware, software, and access issues. Tier 1 handles password resets, software installs, and common errors. Tier 2 tackles network, server, and application issues. Tier 3 and engineering own root cause fixes. Every interaction requires empathy, clear language, and accurate ticket notes.

Code Reference:

```
# Tier 1: intake -> triage -> resolve -> document
# Tier 2: escalate with full context and logs attached
```

What it shows: Support tiers define scope and escalation paths.

Topic guide

What it actually is

IT support is frontline technical assistance for organizational users.

When to use it

When users cannot work due to technology failures.

Where to use it

Corporate helpdesks, schools, hospitals, MSPs, and SaaS internal IT.

Real use example

A teacher cannot open PDF reports; Tier 1 confirms browser cache issue, clears it, and closes the ticket in 12 minutes.

Chapter 2: CompTIA A+ Hardware Fundamentals [Beginner]

Chapter focus: CompTIA A+ Hardware Fundamentals** *(Level: Beginner)

A+ Core 1 covers motherboards, CPU, RAM, storage (SSD/HDD/NVMe), power supplies, and peripherals. Support techs identify failed components, reseal cables, run diagnostics, and order replacements. Understand form factors, POST beep codes, and safe ESD handling.

Code Reference:

```
Get-CimInstance Win32_PhysicalMemory | Select Capacity, Speed  
Get-PhysicalDisk | Select FriendlyName, MediaType, HealthStatus
```

What it shows: PowerShell queries RAM and disk health on Windows endpoints.

Topic guide

What it actually is

Hardware fundamentals are the physical layer of troubleshooting.

When to use it

When devices fail POST, overheat, or show storage errors.

Where to use it

Desktop support, laptop repair, and datacenter walk-throughs.

Real use example

Laptop won't boot; tech reseats RAM stick after beep code chart points to memory failure.

Chapter 3: CompTIA A+ Software and OS Concepts [Beginner]

Chapter focus: CompTIA A+ Software and OS Concepts** *(Level: Beginner)

Core 2 covers Windows editions, macOS basics, Linux distributions, mobile OS, and application

installation. Know 32 vs 64-bit, driver management, Windows Update, Task Manager, and safe mode. Image deployment and user profile migration are common enterprise tasks.

Code Reference:

```
Get-Process | Sort-Object CPU -Descending | Select -First 10  
Get-WmiObject Win32_OperatingSystem | Select Caption, Version
```

What it shows: Identify runaway processes and confirm OS version before patching.

Topic guide

What it actually is

OS concepts explain how software runs on hardware.

When to use it

During malware cleanup, performance issues, and upgrade planning.

Where to use it

All endpoint support roles.

Real use example

PC runs slow; support finds Chrome using 4 GB RAM and advises tab limits plus extension audit.

Chapter 4: Windows Administration for Support Techs [Beginner]

Chapter focus: Windows Administration for Support Techs *(Level: Beginner)**

Support techs reset passwords in Active Directory, manage local admins, check Event Viewer, restart services, and use Settings vs Control Panel. PowerShell cmdlets like Get-Service, Restart-Service, and Get-EventLog speed repetitive tasks.

Code Reference:

```
Get-ADUser -Identity jsmith -Properties PasswordLastSet, LockedOut  
Unlock-ADAccount -Identity jsmith
```

What it shows: Check lockout state and unlock without unnecessary password reset.

Topic guide

What it actually is

Windows admin skills let support fix account and service issues locally and in AD.

When to use it

Daily password, printer, and software ticket work.

Where to use it

Domain-joined corporate laptops and VDI pools.

Real use example

User locked out after vacation; support unlocks account and confirms MFA device is registered.

Chapter 5: Linux Command Line for Support [Beginner]

Chapter focus: Linux Command Line for Support** *(Level: Beginner)

Many servers and cloud VMs run Linux. Support uses ssh, ls, cd, cat, grep, systemctl, journalctl, and tail -f for log review. File permissions (chmod/chown) and package managers (apt, yum) fix common service failures.

Code Reference:

```
ssh admin@server01
sudo systemctl status nginx
sudo journalctl -u nginx --since '1 hour ago'
```

What it shows: Remote session checks web server status and recent error logs.

Topic guide

What it actually is

Linux CLI is essential for server-side support and DevOps handoffs.

When to use it

When applications run on Ubuntu/RHEL in AWS or Azure.

Where to use it

Web app support, database servers, and container hosts.

Real use example

Website down; support ssh in, finds nginx stopped, runs systemctl start nginx, verifies with curl.

Chapter 6: Customer Communication and Soft Skills [Beginner]

Chapter focus: Customer Communication and Soft Skills** *(Level: Beginner)

Technical skill alone is insufficient. Use active listening, avoid jargon with non-technical users, set time expectations, and confirm resolution. Document every step in the ticket so the next shift can continue. De-escalate frustrated callers with empathy and status updates.

Code Reference:

```
# Ticket note template:
# Issue: ... | Steps tried: ... | Resolution: ... | User confirmed: Y/N
```

What it shows: Structured notes improve handoffs and SLA reporting.

Topic guide

What it actually is

Soft skills turn technical fixes into positive user experiences.

When to use it

Every user-facing support interaction.

Where to use it

Phone, chat, walk-up bar, and remote support.

Real use example

Executive upset about email delay; tech explains maintenance window, offers mobile workaround, schedules follow-up.

Track: Intermediate

Chapter 7: Structured Troubleshooting Methodology [Intermediate]**Chapter focus: Structured Troubleshooting Methodology** *(Level: Intermediate)**

Follow identify, establish theory, test, plan, implement, verify, document. CompTIA emphasizes dividing problems (hardware vs software vs network) and checking simplest causes first (cables, reboot, recent changes). Never skip verification with the user.

Code Reference:

1. Reproduce 2. Isolate variable 3. Test hypothesis 4. Verify fix 5. Root cause note

What it shows: Repeatable methodology prevents random guessing.

Topic guide**What it actually is**

Structured troubleshooting is a systematic diagnostic process.

When to use it

Every non-trivial incident.

Where to use it

Tier 1 and Tier 2 daily work.

Real use example

Outlook sync fails only on Wi-Fi; tech disables VPN split tunnel conflict and confirms calendar sync.

Chapter 8: TCP/IP, DNS, and DHCP Essentials [Intermediate]**Chapter focus: TCP/IP, DNS, and DHCP Essentials** *(Level: Intermediate)**

Understand IPv4 addressing, subnet masks, default gateway, and DHCP lease renewal. DNS translates names to IPs; stale DNS causes 'site not found' after migrations. Tools: ipconfig /all, ping, tracert, nslookup, and Test-NetConnection.

Code Reference:

```
ipconfig /all
nslookup codequest.app
Test-NetConnection codequest.app -Port 443
```

What it shows: Full stack check from IP config through DNS to HTTPS port.

Topic guide

What it actually is

Networking fundamentals explain most connectivity tickets.

When to use it

Cannot reach internal apps, websites, or VPN resources.

Where to use it

Remote work, office LAN, and hybrid cloud access.

Real use example

After datacenter move, users hit old IP; flushing DNS and verifying A record fixes access.

Chapter 9: Remote Support Tools and VPN [Intermediate]

Chapter focus: Remote Support Tools and VPN** *(Level: Intermediate)

Remote tools include Quick Assist, TeamViewer, RDP, and SSH. VPN connects users to corporate networks; split tunnel vs full tunnel affects performance and security. Always obtain user consent before remote control.

Code Reference:

```
mstsc /v:workstation-42  
# Or: ssh -L 8080:localhost:80 jump-host
```

What it shows: RDP for Windows; SSH tunnel for secure port forwarding.

Topic guide

What it actually is

Remote support enables fix-without-truck-roll.

When to use it

Work-from-home and branch office scenarios.

Where to use it

Global enterprises and MSP remote desks.

Real use example

Field sales rep cannot reach CRM; support confirms VPN client outdated, pushes new profile.

Chapter 10: Ticketing Systems and SLA Management [Intermediate]

Chapter focus: Ticketing Systems and SLA Management** *(Level: Intermediate)

Tickets track priority (P1-P4), category, assignment group, and SLA clocks (response vs resolution). Breached SLAs escalate automatically. Good categorization enables trend reporting

and problem management.

Code Reference:

```
Priority = f(impact, urgency)
P1: many users / critical app down
P4: single user / low impact
```

What it shows: Impact × urgency matrix drives priority.

Topic guide

What it actually is

Ticketing is the system of record for IT support work.

When to use it

All professional IT organizations.

Where to use it

ServiceNow, Jira SM, Zendesk, Freshservice.

Real use example

Email outage for 200 users logged P1; SLA page fires on-call engineer within 15 minutes.

Chapter 11: Active Directory and Account Management [Intermediate]

Chapter focus: Active Directory and Account Management *(Level: Intermediate)**

AD stores users, groups, computers, and Group Policy. Support creates/disables accounts, resets passwords, adds group membership, and troubleshoots GPO application with gpresult. Azure AD Connect syncs hybrid identities.

Code Reference:

```
Get-ADGroupMember 'VPN-Users'
gpresult /R /SCOPE USER
```

What it shows: Verify group membership and applied GPO on user login.

Topic guide

What it actually is

Active Directory is the identity backbone of Windows enterprises.

When to use it

Onboarding, offboarding, and access issues.

Where to use it

Hybrid Microsoft 365 environments.

Real use example

New hire missing SharePoint access; support adds to SG-Finance-Users, user logs off/on, access works.

Chapter 12: ITIL Service Management Basics [Intermediate]

Chapter focus: ITIL Service Management Basics** *(Level: Intermediate)

ITIL 4 defines value co-creation through services. Key practices: incident (restore service), problem (root cause), change (controlled modification), and request fulfillment. Service catalog lists standard offerings; CMDB supports impact analysis.

Code Reference:

```
Incident -> (workaround) -> Problem -> Known Error -> Change -> Release
```

What it shows: Practice chain from firefight to permanent fix.

Topic guide

What it actually is

ITIL provides common language for IT service delivery.

When to use it

Mature IT departments aligning support with business outcomes.

Where to use it

Enterprises, government, and regulated industries.

Real use example

Recurring printer driver crash becomes problem record; known error published until patch change.

Chapter 13: Security Basics for Support Staff [Intermediate]

Chapter focus: Security Basics for Support Staff** *(Level: Intermediate)

Support is a phishing target with elevated credentials. Verify caller identity before password resets. Recognize social engineering. Enforce MFA enrollment. Never share admin passwords. Report suspicious emails to SOC.

Code Reference:

```
# Before reset: verify employee ID + manager approval for VIP accounts
```

What it shows: Identity verification prevents account takeover via fake support calls.

Topic guide

What it actually is

Security-aware support reduces insider-threat and social-engineering risk.

When to use it

Every password reset and remote session.

Where to use it

Helpdesks with AD and privileged tool access.

Real use example

Attacker impersonates CFO; tech follows callback procedure and blocks attempt.

Track: Advanced

Chapter 14: PowerShell Automation for Helpdesk [Advanced]**Chapter focus: PowerShell Automation for Helpdesk** *(Level: Advanced)**

Automate onboarding: create AD user, assign groups, create mailbox, add license. Scheduled scripts check disk space, stale accounts, and expired passwords. Use approved script repository and logging.

Code Reference:

```
$users = Import-Csv new_hires.csv
foreach ($u in $users) {
    New-ADUser -Name $u.Name -SamAccountName $u.Sam -Enabled $true
}
```

What it shows: Bulk user creation from HR feed reduces manual errors.

Topic guide**What it actually is**

PowerShell automation scales Tier 1 repetitive tasks.

When to use it

Onboarding batches, health checks, and reporting.

Where to use it

Microsoft-centric enterprises.

Real use example

Monday onboarding script creates 15 users in 3 minutes vs 2 hours manual work.

Chapter 15: Enterprise Monitoring and Asset Management [Advanced]**Chapter focus: Enterprise Monitoring and Asset Management** *(Level: Advanced)**

Tools like SCCM/Intune, Lansweeper, or Snipe-IT track hardware lifecycle, warranty, and patch compliance. Integrate alerts with ticketing for proactive fixes before users call.

Code Reference:

```
# Intune compliance policy: BitLocker ON, OS patch < 30 days, AV active
```

What it shows: Compliance policies define healthy endpoint baseline.

Topic guide

What it actually is

Monitoring shifts support from reactive to proactive.

When to use it

Large fleets (1000+ endpoints).

Where to use it

Enterprise desktop and mobile management.

Real use example

Disk space alert opens ticket before user crashes during exam submission.

Chapter 16: Disaster Recovery and Backup Verification [Advanced]

Chapter focus: Disaster Recovery and Backup Verification *(Level: Advanced)**

Support participates in DR tests: verify backup restore, document RTO/RPO, and run tabletop exercises. Know offsite backup location and escalation during ransomware events.

Code Reference:

```
Restore-ADObject -Identity $guid -Server DC1  
# Test file restore from Veeam backup job
```

What it shows: Periodic restore tests prove backups are not vanity metrics.

Topic guide**What it actually is**

DR verification ensures recoverability after catastrophic failure.

When to use it

Quarterly DR drills and post-incident recovery.

Where to use it

Datacenters, schools, and healthcare IT.

Real use example

Ransomware hits file server; team restores from immutable backup within 4-hour RTO.

Chapter 17: Tier 3 Escalation and Root Cause Analysis [Advanced]

Chapter focus: Tier 3 Escalation and Root Cause Analysis *(Level: Advanced)**

Tier 3 uses deep logs, perfmon, Process Monitor, and vendor support cases. Apply 5 Whys and fishbone diagrams. Produce RCA documents with corrective and preventive actions.

Code Reference:

```
Why 1: App slow -> DB timeout  
Why 2: Pool exhausted -> ...  
Why 5: Missing index on query
```

What it shows: 5 Whys trace symptoms to systemic cause.

Topic guide

What it actually is

RCA prevents repeat incidents and informs problem management.

When to use it

Major incidents and recurring escalations.

Where to use it

Engineering and operations bridge roles.

Real use example

Monthly login spike traced to cron job hammering auth service; schedule adjusted.

Chapter 18: Knowledge Base Engineering [Advanced]

Chapter focus: Knowledge Base Engineering** *(Level: Advanced)

Transform ticket resolutions into searchable KB articles with symptoms, environment, steps, and verification. Measure deflection rate and article usefulness scores. Maintain ownership and review cadence.

Code Reference:

KB Template: Title | Symptoms | Environment | Resolution Steps | Related Articles

What it shows: Standard template ensures findability and consistency.

Topic guide

What it actually is

KB engineering reduces ticket volume through self-service.

When to use it

Mature service desks with self-service portals.

Where to use it

ServiceNow Knowledge, Confluence, SharePoint.

Real use example

Password reset KB deflects 40% of Tier 1 tickets after portal promotion.